

Annual control work, and the Grade 8 preview

The whole year in one paper — and what waits in Grade 8.

LESSON 169–170

2 × 40 MIN

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S8 REVIEW

LESSON CLOCK

3

45'

12'

15'

5'

Instructions	3 min
The paper	45 min
Answers	12 min
Diagnosis	15 min
Grade 8	5 min

LEARNING OBJECTIVES

- Work accurately across the whole year under time.
- Choose the right method without being told which chapter a question comes from.
- Diagnose the year's remaining weaknesses.
- See how each block of Grade 7 continues in Grade 8.

TEXTBOOK

National	the whole textbook
Cambridge	Stage 8, whole book
Workbook	Annual paper

01

Explanation

The paper — 40 marks, 45 minutes

Q	TASK	MARKS	QUARTER
1	Compute $-7 + 12 - (-5)$ and $\frac{2}{3} - \frac{1}{4}$	4	I
2	Expand $(3x - 2)^2$ and factorise $x^2 - 49$	5	II–III
3	Simplify $\frac{x^2 - 9}{x + 3}$ and state the excluded value	4	III
4	Solve $3(x - 2) = 2x + 5$	4	III
5	Two parallel lines are cut by a transversal; one angle is 115° : find the other seven	5	III
6	The angles of a triangle are in the ratio $2 : 3 : 4$: find them	4	III
7	Draw $y = 2x - 4$ and give both intercepts	5	IV
8	Solve $\{ x + y = 9 ; x - y = 3 \}$	5	IV
9	How many 3-figure codes from 10 digits have all digits different?	4	IV

NINE QUESTIONS, FOUR QUARTERS

Nothing is asked that has not been taught, and nothing is labelled with its chapter. Recognising which method a question wants is itself the skill being tested.

The answers

Q	ANSWER	THE MARK MOST OFTEN LOST
1	10	the double minus, and the common denominator
2	$9x^2 - 12x + 4$	the middle term, and squaring the 3
3	$x - 3$	$x \neq -3$
4	$x = 11$	expanding the bracket
5	115°	naming which pair is which
6	$40^\circ, 60^\circ, 80^\circ$	$9x = 180$
7	$(0; -4)$	using a ruler
8	$(6; 3)$	reporting both numbers
9	720	$10 \cdot 9 \cdot 8$

Diagnosis

IF YOU LOST MARKS ON	REVISE
Q1	integers and rational numbers — Quarter I
Q2	the abridged multiplication formulae — L98–109
Q3	algebraic fractions — L112–119
Q4	linear equations — L123–126
Q5	angles on parallel lines — L82–85
Q6	the angle sum of a triangle — L86–87
Q7	the linear function — L141–142
Q8	systems — L147–149
Q9	combinatorics — L155–163

REVISE THE BLOCK, NOT THE QUESTION

A lost mark is a symptom. Reworking one question fixes that question; rereading the block fixes the next twenty.

What Grade 8 expects

GRADE 7 BLOCK	GRADE 8 CONTINUES WITH
rational numbers	irrational numbers and square roots
abridged formulae	quadratic equations and completing the square
algebraic fractions	rational expressions and fractional equations
linear equations	inequalities and their intervals
parallel lines and triangles	quadrilaterals, Pythagoras and trigonometry
the linear function	$y = x^2$
systems	systems with a quadratic equation in them
combinatorics	probability from counted outcomes

NOTHING IS DROPPED

Every block above continues; none of it is replaced. The year's work is the ground Grade 8 stands on, which is why the diagnosis above is worth acting on over the summer.

02 Worked examples

EXAMPLE 1 Model answer, Q4: solve $3(x - 2) = 2x + 5$.

- ① $3x - 6 = 2x + 5$
Expand first.

- ② $3x - 2x = 5 + 6$
Both terms change sign.

- ③ $x = 11$

- ④ Check: $3 \cdot 9 = 27$ and $22 + 5 = 27$ ✓

ANSWER

$$x = 11$$

EXAMPLE 2 Model answer, Q5: $a \parallel b$ and one angle is 115° .

- ① Vertical to it: 115° .

- ② Supplementary to it: 65° .

- ③ Corresponding and alternate: 115° .

- ④ Four angles are 115° and four are 65° .

ANSWER

Four of 115° , four of 65°

EXAMPLE 3 Model answer, Q8: $\{x + y = 9; x - y = 3\}$.

- 1 Add the equations: $2x = 12$.
Opposite coefficients of y .
- 2 $x = 6$
- 3 $6 + y = 9$, so $y = 3$.
- 4 Check both ✓ — the answer is the pair $(6; 3)$.

ANSWER

 $(6; 3)$ **03** Interactive model

Give back the diagnosis table with each script and ask every pupil to name their own two weakest blocks; they arrive in Grade 8 knowing where to look.

The year in twelve questions

$-7 + 12 - (-5)$ equals:

0

10

-10

24

$\frac{2}{3} - \frac{1}{4}$ equals:

$\frac{1}{12}$

$\frac{5}{12}$

$\frac{1}{7}$

$\frac{3}{4}$

$(3x - 2)^2$ equals:

$9x^2 + 4$

$9x^2 - 12x + 4$

$3x^2 - 12x + 4$

$9x^2 - 6x + 4$

$x^2 - 49$ factorises to:

$x^2 - 9$ $x + 3$ equals:

$3(x - 2) = 2x + 5$ gives:

On parallel lines, an angle of 115° has co-interior angle:

Angles in the ratio 2 : 3 : 4 are:

$40^\circ, 60^\circ, 80^\circ$

$30^\circ, 60^\circ, 90^\circ$

$20^\circ, 30^\circ, 40^\circ$

$45^\circ, 60^\circ, 75^\circ$

$y = 2x - 4$ cuts Ox at:

$(0; -4)$

$(2; 0)$

$(4; 0)$

$(-2; 0)$

$\{ x + y = 9 ; x - y = 3 \}$ gives:

$(6; 3)$

$(3; 6)$

$(9; 3)$

$(4; 5)$

3-figure codes from 10 digits, all different:

1000

720

120

30

Handshakes among 10 people:

45

90

100

20

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Annual control work	Yillik nazorat ishi	Годовая контрольная работа
Revision	Takrorlash	Повторение
Method	Usul	Метод
Accuracy	Aniqlik	Точность
Diagnosis	Tashxis	Диагностика
Preview	Oldindan tanishtirish	Предварительный обзор
Progress	Rivojlanish	Прогресс
Grade 8	8-sinf	8 класс

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Q1 tests:

integers and fractions

equations

geometry

counting

Q2 tests:

fractions

the abridged formulae

systems

probability

Q3 requires:

a graph

an excluded value

a construction

a ratio

Q5 tests:

triangles

angles on parallel lines

constructions

coordinates

Q8 must be answered with:

one number

an ordered pair

a graph

an equation

Q9 uses:

the addition rule

the multiplication rule without repetition

a ratio

a proportion

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $-7 + 12 - (-5)$

ANSWER 10

2. $\frac{2}{3} - \frac{1}{4}$

ANSWER $\frac{5}{12}$

3. $(3x - 2)^2$

ANSWER $9x^2 - 12x + 4$

4. $x^2 - 49$

ANSWER $(x - 7)(x + 7)$

5. $\frac{x^2 - 9}{x + 3}$

ANSWER $x - 3$

6. $3(x - 2) = 2x + 5$

ANSWER $x = 11$

7. Angles in the ratio 2 : 3 : 4

ANSWER $40^\circ, 60^\circ, 80^\circ$

Medium · the standard question

7 PROBLEMS

1. On parallel lines, the co-interior angle of 115°

ANSWER 65°

2. $y = 2x - 4$: both intercepts

ANSWER $(0; -4)$ and $(2; 0)$

3. $\{x + y = 9; x - y = 3\}$

ANSWER $(6; 3)$

4. 3-figure codes, all digits different

ANSWER 720

5. Handshakes among 10 people

ANSWER 45

6. A cuboid $5 \times 4 \times 3$: its volume

ANSWER 60 cm^3

7. Share 120 in $2 : 3$

ANSWER 48 and 72

Hard · stretch and reason

7 PROBLEMS

1. Factorise $2x^2 - 18$ completely

ANSWER $2(x - 3)(x + 3)$

2. Simplify $\frac{1}{x - 1} - \frac{1}{x + 1}$

ANSWER $\frac{2}{x^2 - 1}$

3. Solve $\{ 2x + 3y = 12 ; x - y = 1 \}$

ANSWER $(3; 2)$

4. An exterior angle of 125° , one remote angle 47°

ANSWER 78°

5. A right triangle with hypotenuse 16 and an angle of 30°

ANSWER The opposite leg is 8

6. How many 3-figure codes contain at least one 7?

ANSWER 271

7. Two numbers add to 26 and differ by 8

ANSWER 17 and 9

07 Homework

Over the summer — 5 tasks

Two blocks, chosen from your own diagnosis, are worth more than all eight done lightly.

1. Name your two weakest blocks from the diagnosis table.
2. Rework every question of the annual paper you lost a mark on.
3. Write out the five abridged multiplication formulae from memory and check them.
4. Solve ten linear equations and ten systems from the textbook exercises.

5. Read ahead: find out what an irrational number is, and bring one example to the first Grade 8 lesson.